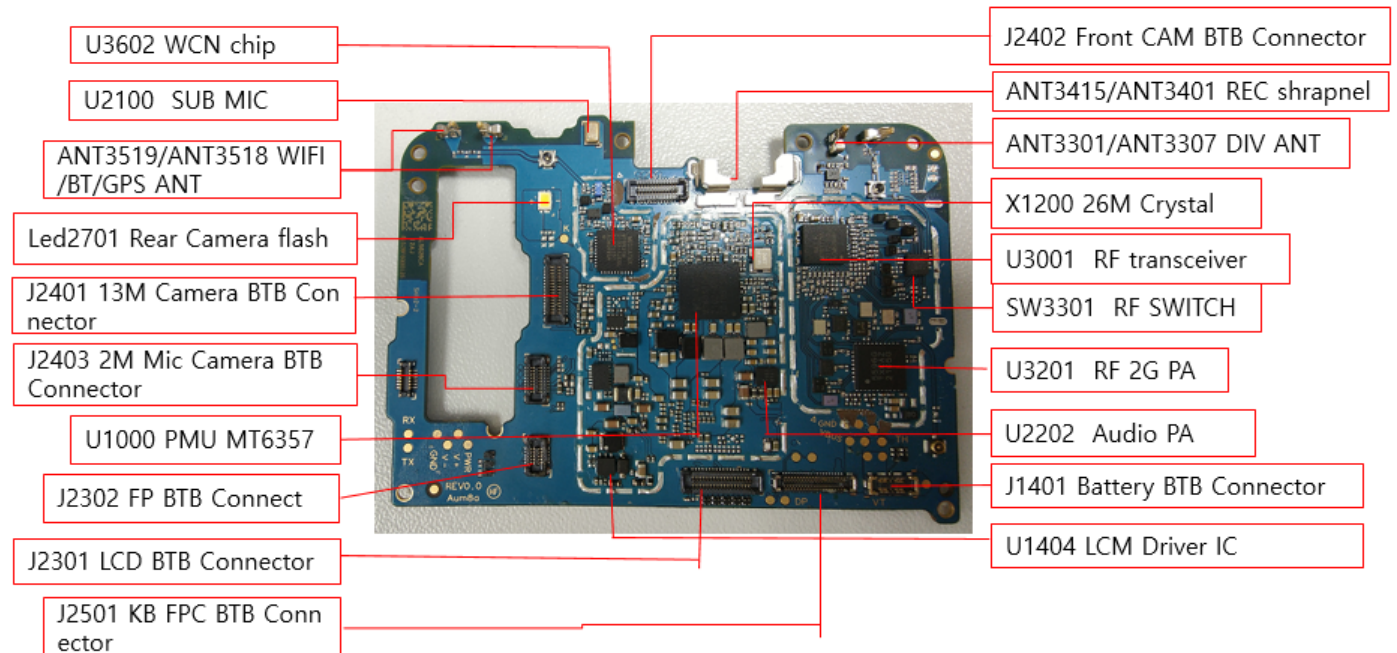


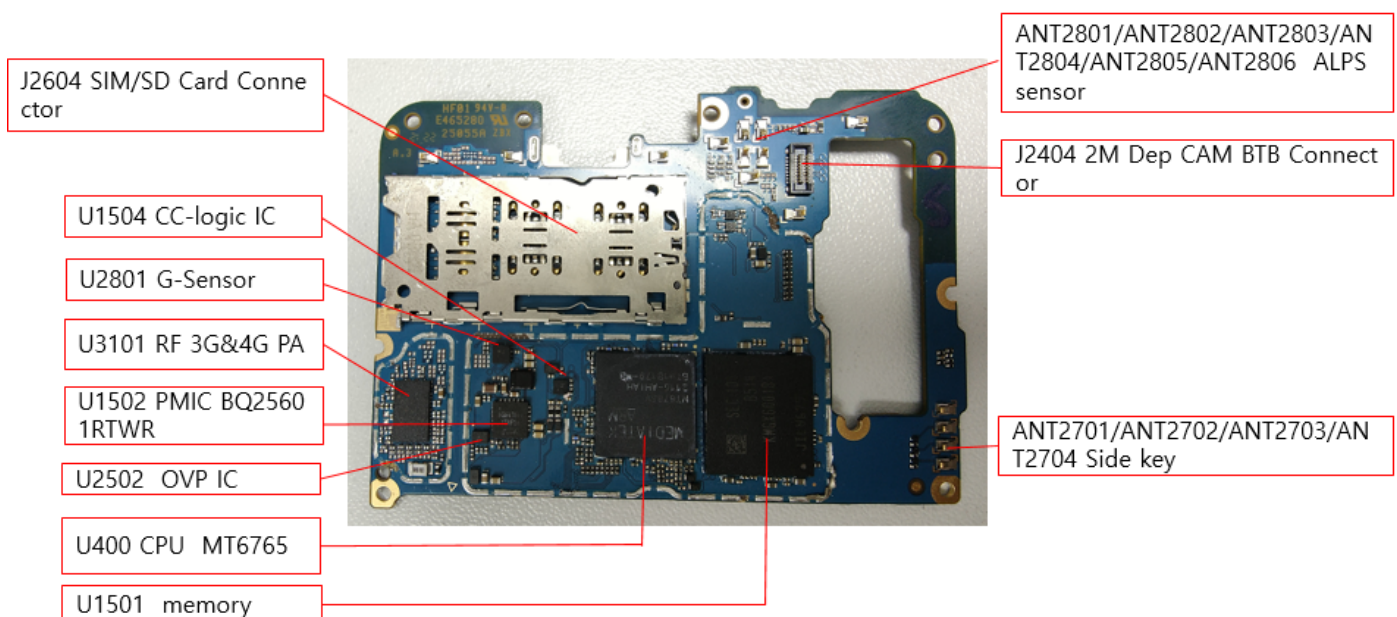
8. Level 3 Repair

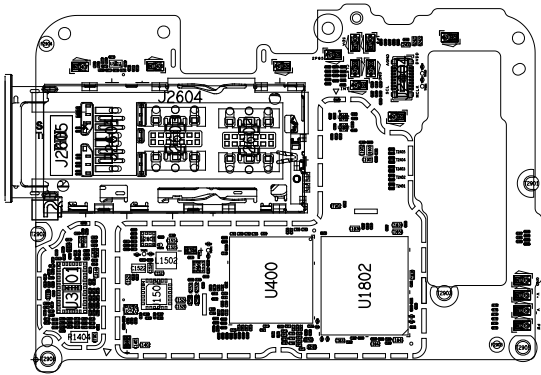
8-1. Components Layout

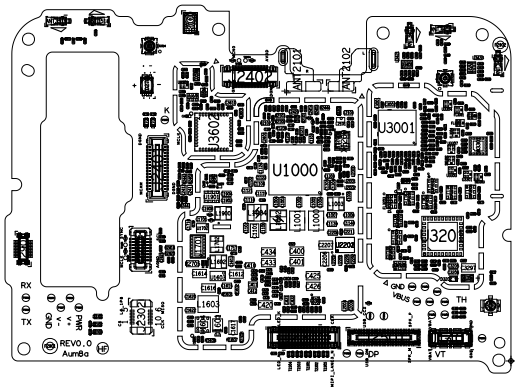
PBA (Bottom)

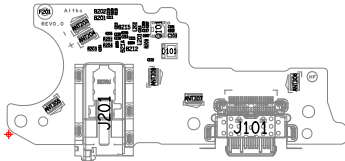


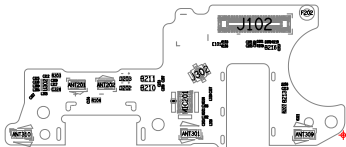
PBA (TOP)



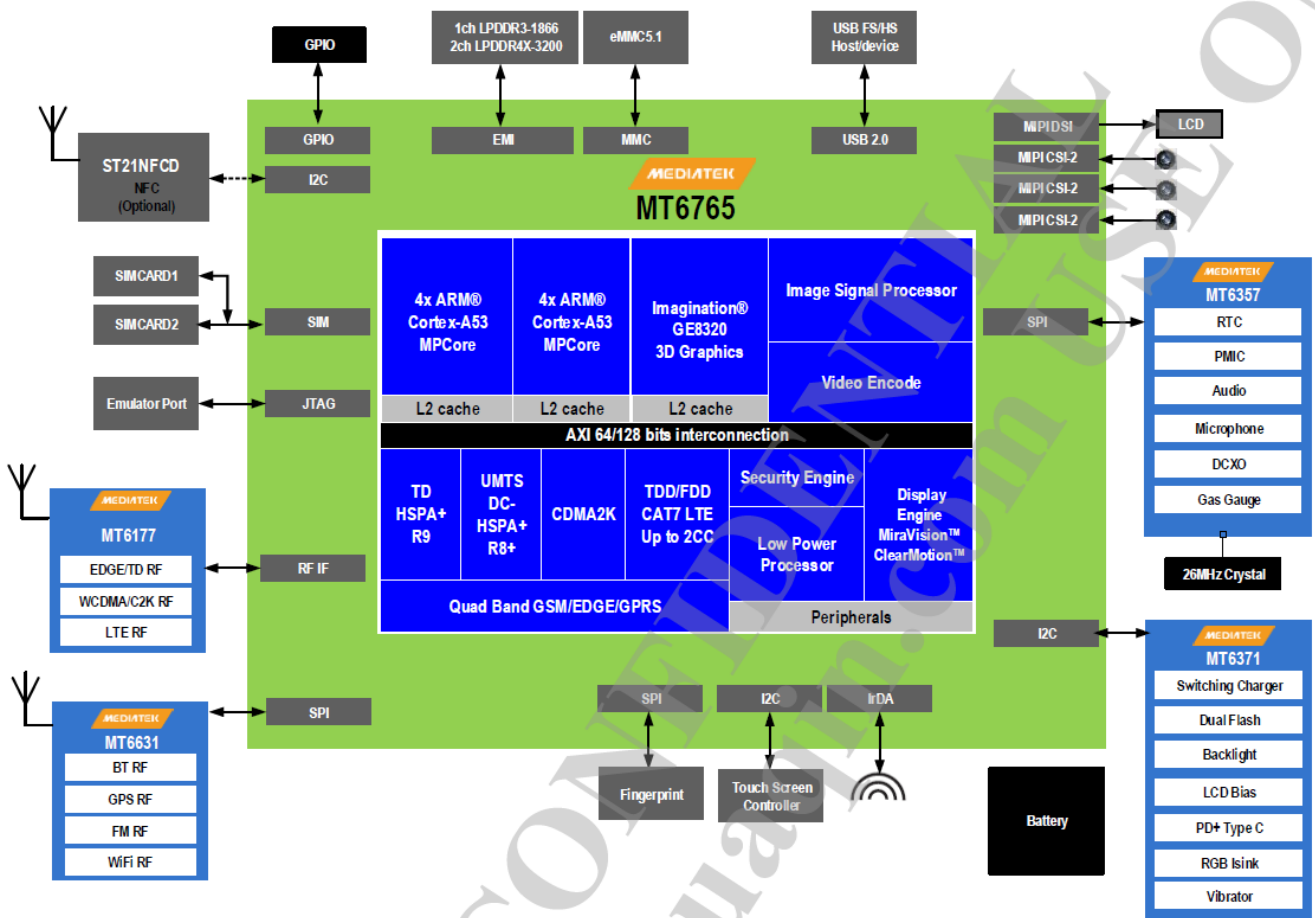




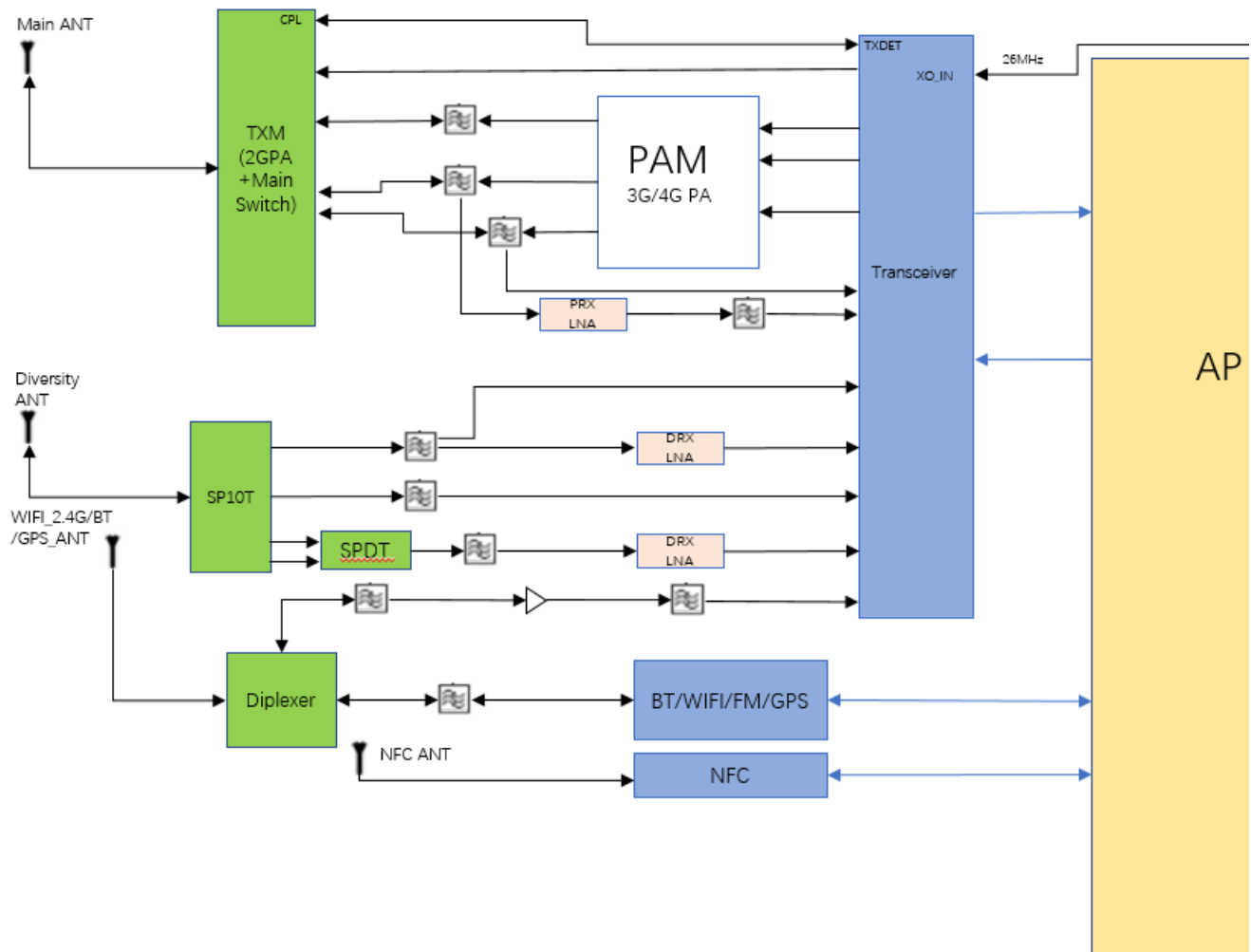




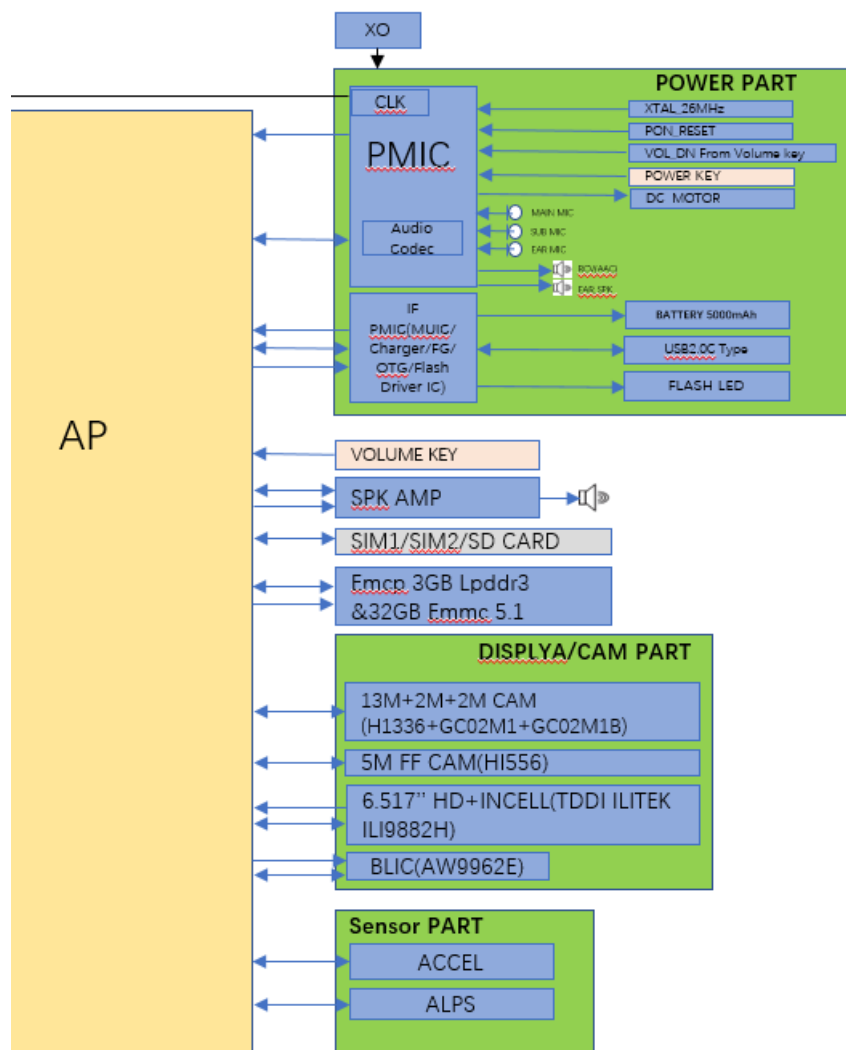
8. Level 3 Repair



8. Level 3 Repair

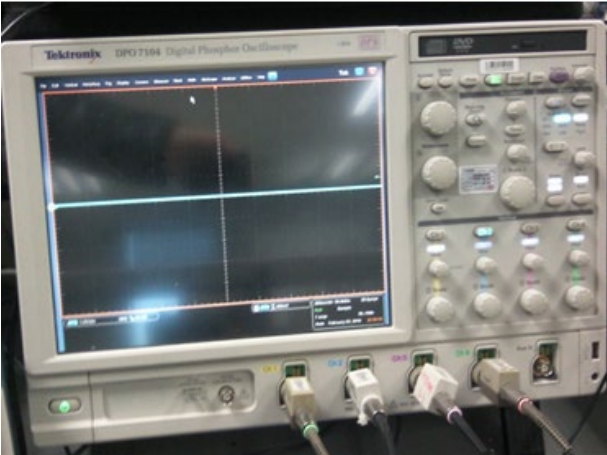
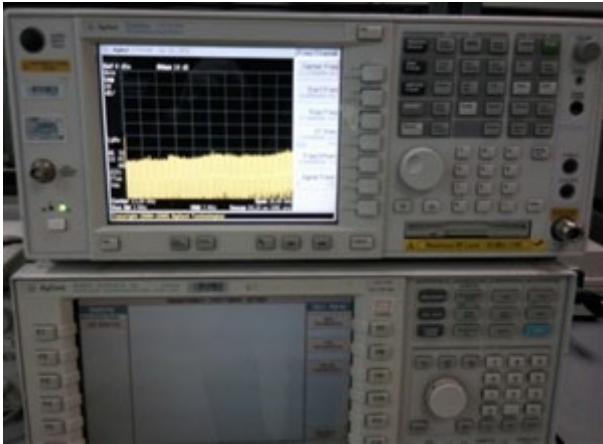


8. Level 3 Repair



8. Level 3 Repair

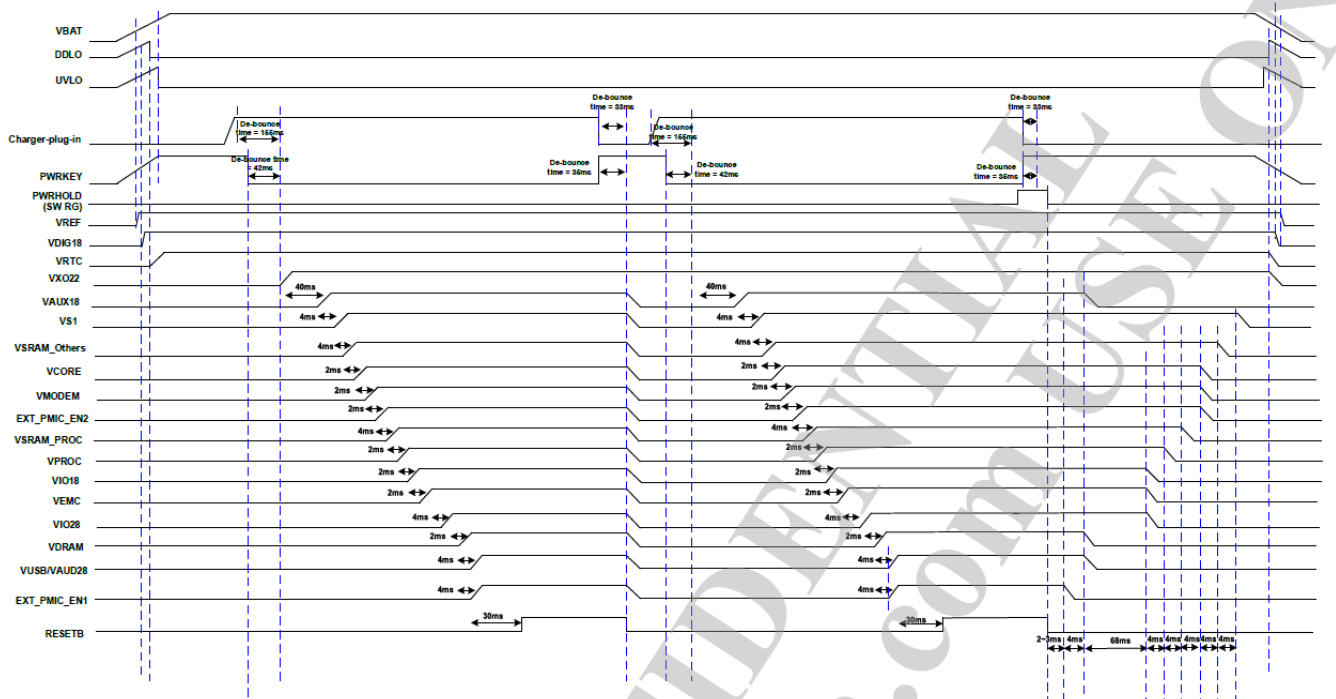
8-2. Flow chart of Troubleshooting.

	
Oscilloscope	Digital Multimeter
	
Power Supply	+ driver, ESD Safe Tweezer
	
8960 & Spectrum Analyzer	Soldering iron

8. Level 3 Repair

8-3-1. Power On

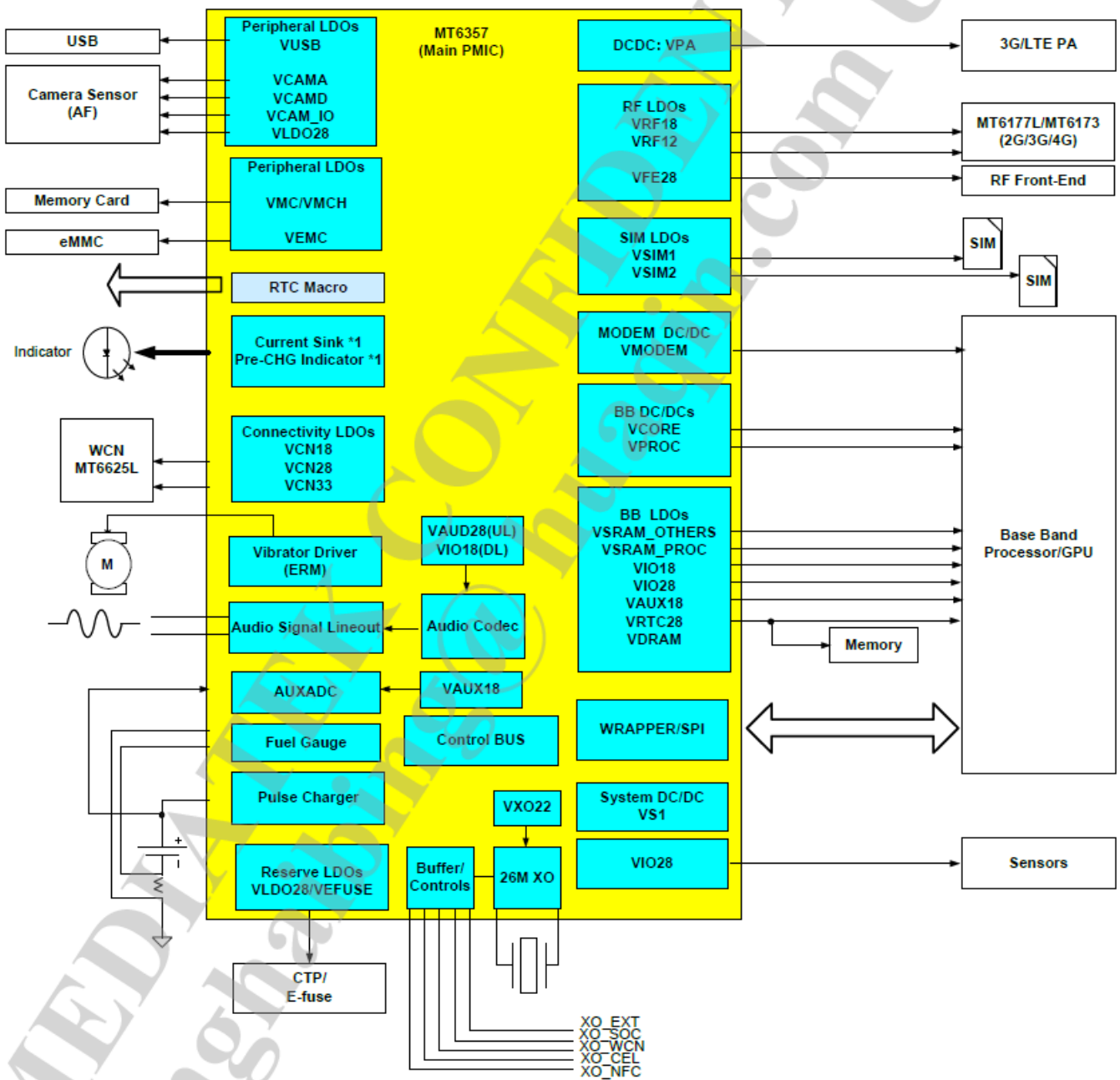
■ Checking Power signal (Battery connector, PMU, Clock)



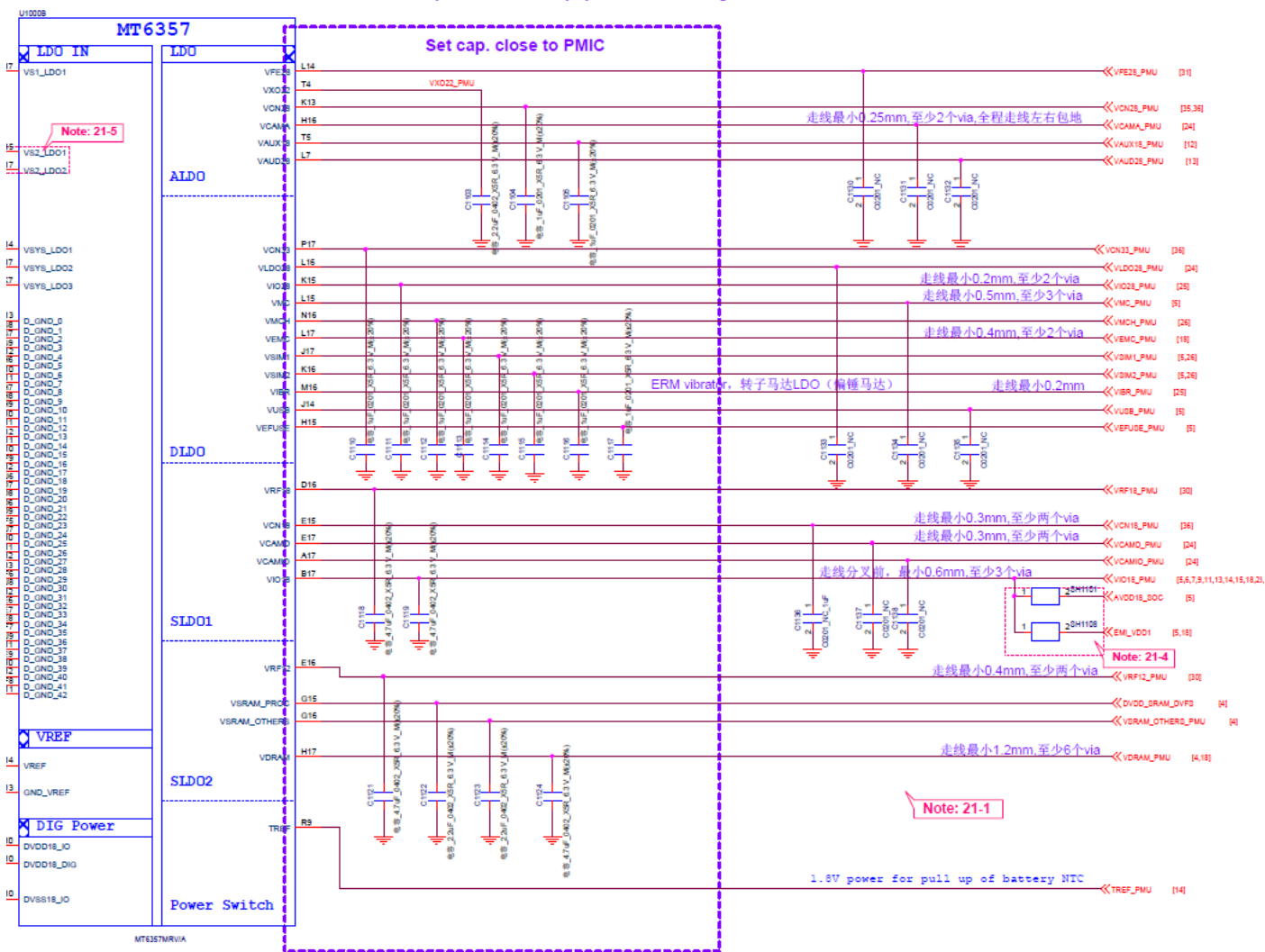
Note: Those timings are typical values; timing variation is $\pm 20\%$ (Charger-plug-in/out related delay timing variation is $\pm 40\%$).

Start up voltage sequence diagram

8. Level 3 Repair



8. Level 3 Repair



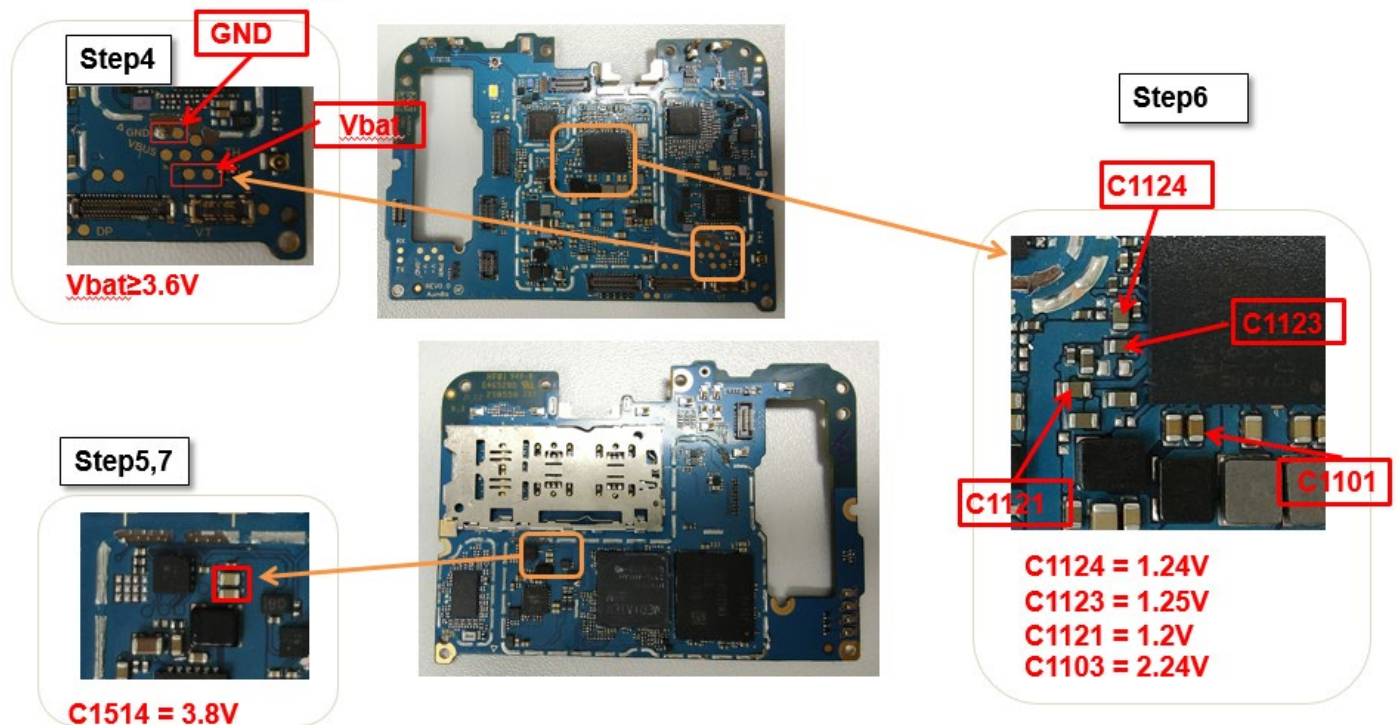
VREG_LDO schematic diagram

8. Level 3 Repair

No Power

Step	Check point	Result value	Defect point
1	Confirm the defect symptom. (Do not confuse the screen problem.)	-	-
2	Check the Power button working physically	Normal	Go to the step 3
		Get Stuck / Sticky	Cleaning, Reassembly
3	Can USB charge normally ?	Yes	Go to the step 3.1
		No	Replace the battery
3.1	Can start the machine normally after charging ?	Yes	Cell phone under voltage, c ontinue charging
		No	Go to the step 4
4	Check the voltage level at Vbat and GND these t wo points is greater than 3.8V	Yes	Go to the step 5
		No	Replace the battery
5	Check whether the voltage at both ends of c15 14 capacitor is greater than 3.8V	Yes	Go to the step 6
		No	Go to the step 7
6	Check the output voltage of U1000, C1123=1.25 V, C1121=1.2V, C1124=1.24V, C1103=2.24V	If the output voltage is not normal, Change the U1000	
7	Check the output voltage of U1502. C1514=3.8V(Consider voltage margin $\pm 10\%$)	If the output voltage is not normal, Change the U1502	

No Power (cont')



8. Level 3 Repair

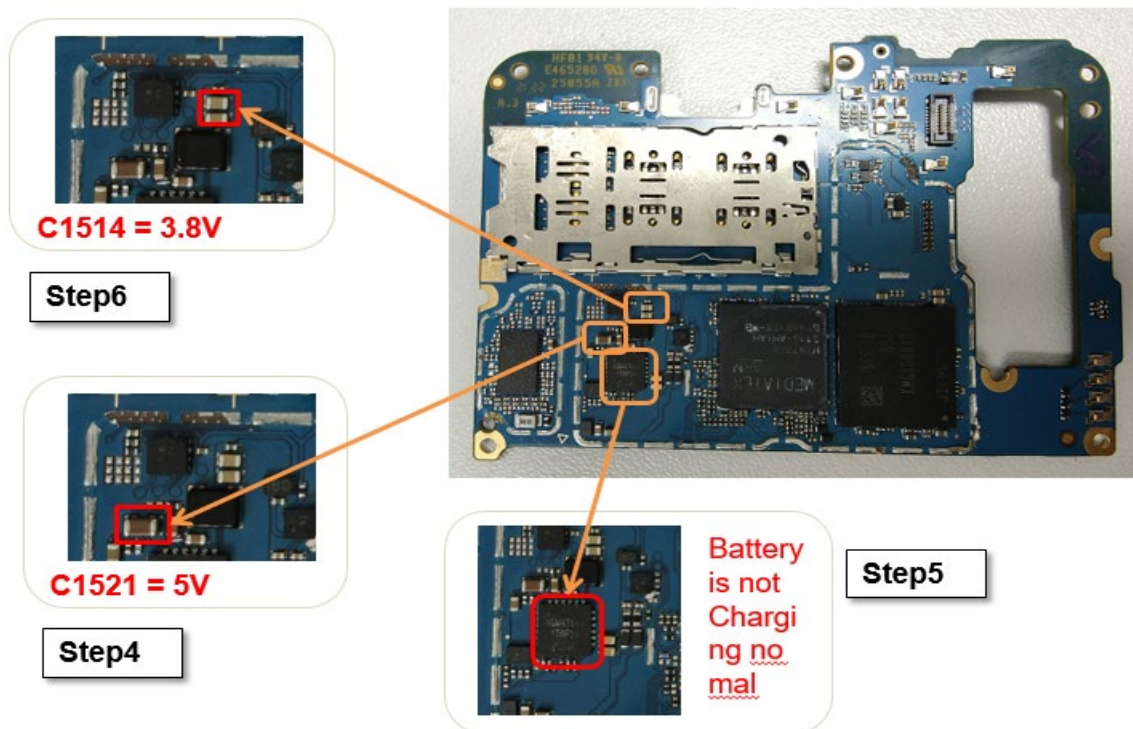
8-3-2. Charging

- The charging controlled by PMU chip MT6357 (U1000) , PMIC chip (U1502) and OVP chip U2502,

No Charging

Step	Check point	Result value	Defect point
1	Confirm the defect symptom	-	-
2	Replace a battery.	Solved	Go to the step 2.1
		Not solved	Go to the step 3
2.1	Charge the customer battery during 5min at least.	Solved	Totally discharged battery
		Not solved	Go to the step 3
3	Disassemble and check I/F connector visually	Dust	Clean I/F connector
		Damage	Replace I/F connector
		Normal	Go to the step 4
4	Check the VBUS_USB_IN(C1521)=5V	C1521 = 5V	Go to the step 6
		If not the correct value	Replace U2502
5	Battery is Charging <u>normal</u>	No	Replace U1502
6	Check whether the voltage at both ends of c1514 capacitor is greater than 3.8V	If the output voltage is not normal, Change the U1502	

No Charging

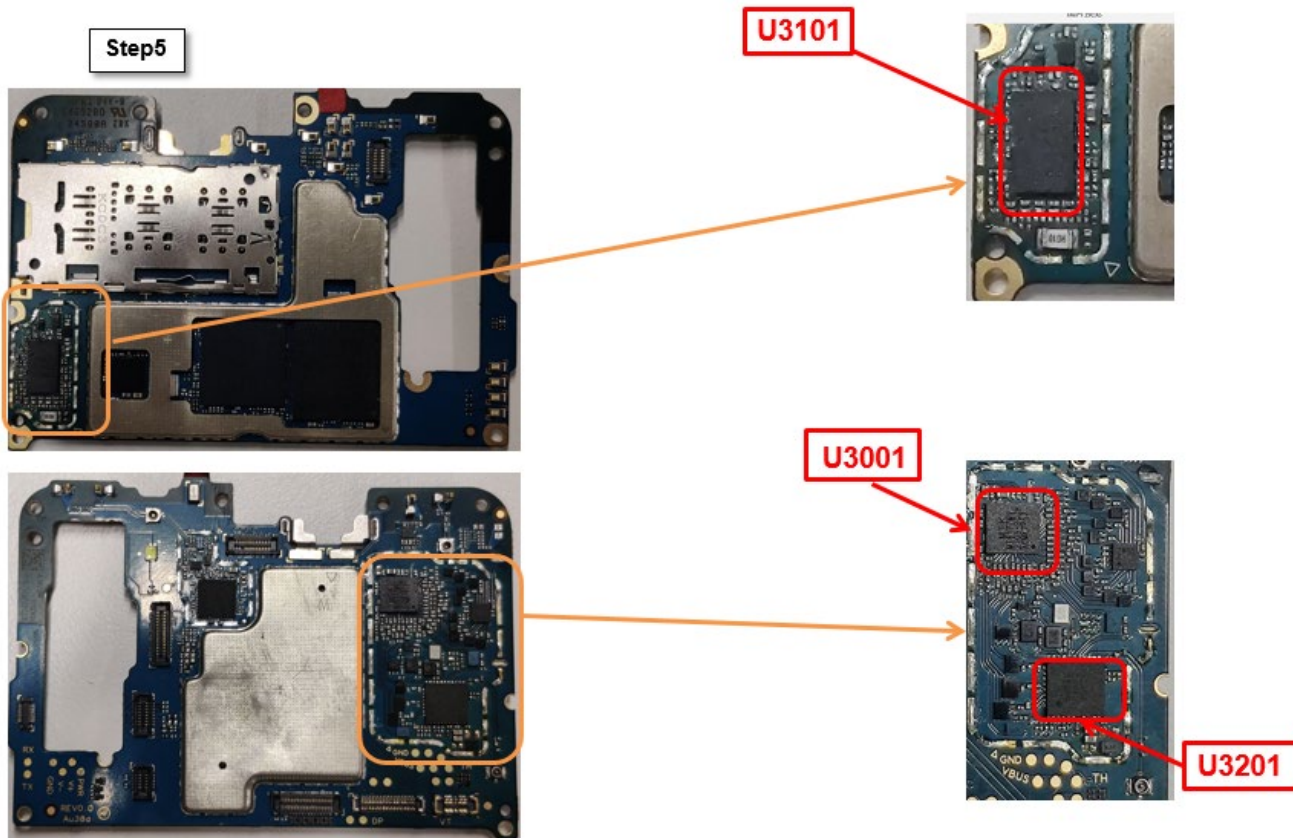


8. Level 3 Repair

Call Problem (with RF test equipment)

Step	Check point	Result value		Defect point
1	Confirm the defect symptom	-		-
2	RF radiation test	Pass		Network or Settings
		Fail		Go to the next step
3	RF calibration	Pass		Go to the step 4
		Fail		Go to the step 5
4	RF radiation test	Pass		Repaired
		Fail		Except PBA (Coaxial cable, Antenna, Shielding condition)
5	A type of failure	TX	2G	TXM(U3201) TRANSCEIVER(U3001)
			3G/LTE	PA(U3101) TXM(U3201) TRANSCEIVER(U3001)
		RX	2G	TRANSCEIVER(U3001) TXM(U3201)
			3G/LTE	TRANSCEIVER(U3001) TXM(U3201)

Call Problem



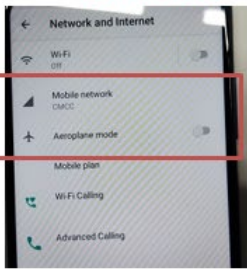
8. Level 3 Repair

Call Problem (without RF test equipment)

Step	Check point	Result value	Defect point
1	Confirm the defect symptom	-	-
2	Check the settings (airplane mode, Mobile networks)	Abnormal	Settings
		Normal	Go to the next step
3	Check the signal strength of the device <u>Settings->About phone->SIM status->Signal strength.</u> (Compare to normal device)	Abnormal	Go to the next step
		Normal	Network
4	Check the RF parts except PBA. (Coaxial cable, Antenna, Shielding condition, etc..)	Broken, dust, corrosion	RF parts
		Loose fitting	Connection
		Normal	Go to the next step
5	Check the status visually(crack, missing, Corrosion..etc) of RF components. (compare to normal PBA) TRANSCEIVER(U3001) TXM(U3201) PA(U3101) DUP3201,DUP3203,DUP3204 etc.	Abnormal	MMMB(PAM1000) TRANSCEIVER(U2012)
		Normal	CP(Call Processor) (U400) PMIC(U1501)

Call Problem

Step2,3 CHECK SETTINGS



Step4



Step4

Diversity ANTENNA contact



Step4

Coaxial cable (connect Main Ant. with PBA)



ANT. Contact (for main RF)

Step5

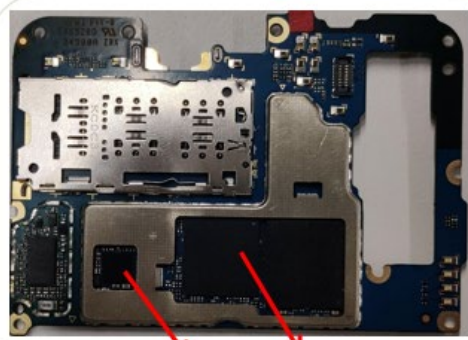


Signal strength
Same value compare to normal device.

8. Level 3 Repair

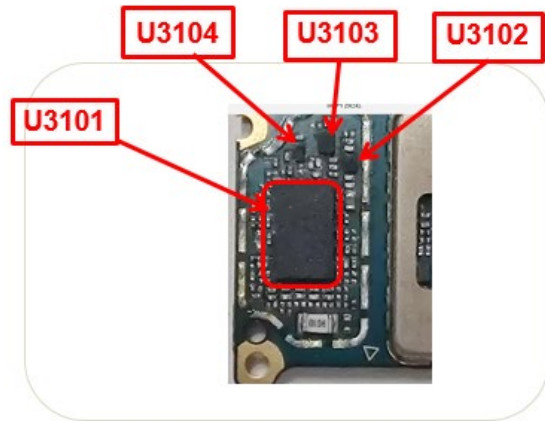
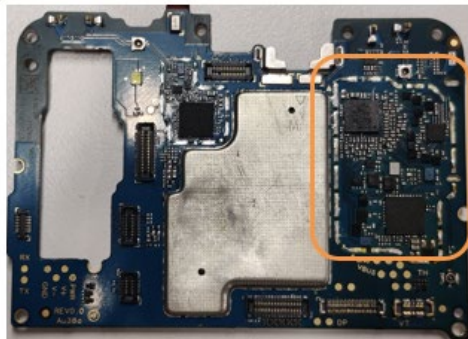
Call Problem

Step5



U1501

U400

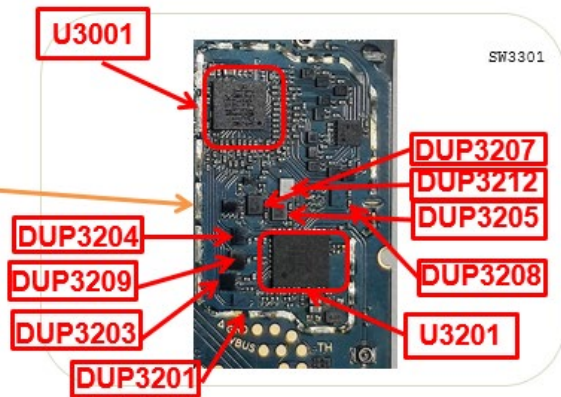


U3101

U3104

U3103

U3102



U3001

DUP3204

DUP3209

DUP3203

DUP3201

DUP3207

DUP3212

DUP3205

DUP3208

U3201

SW3301

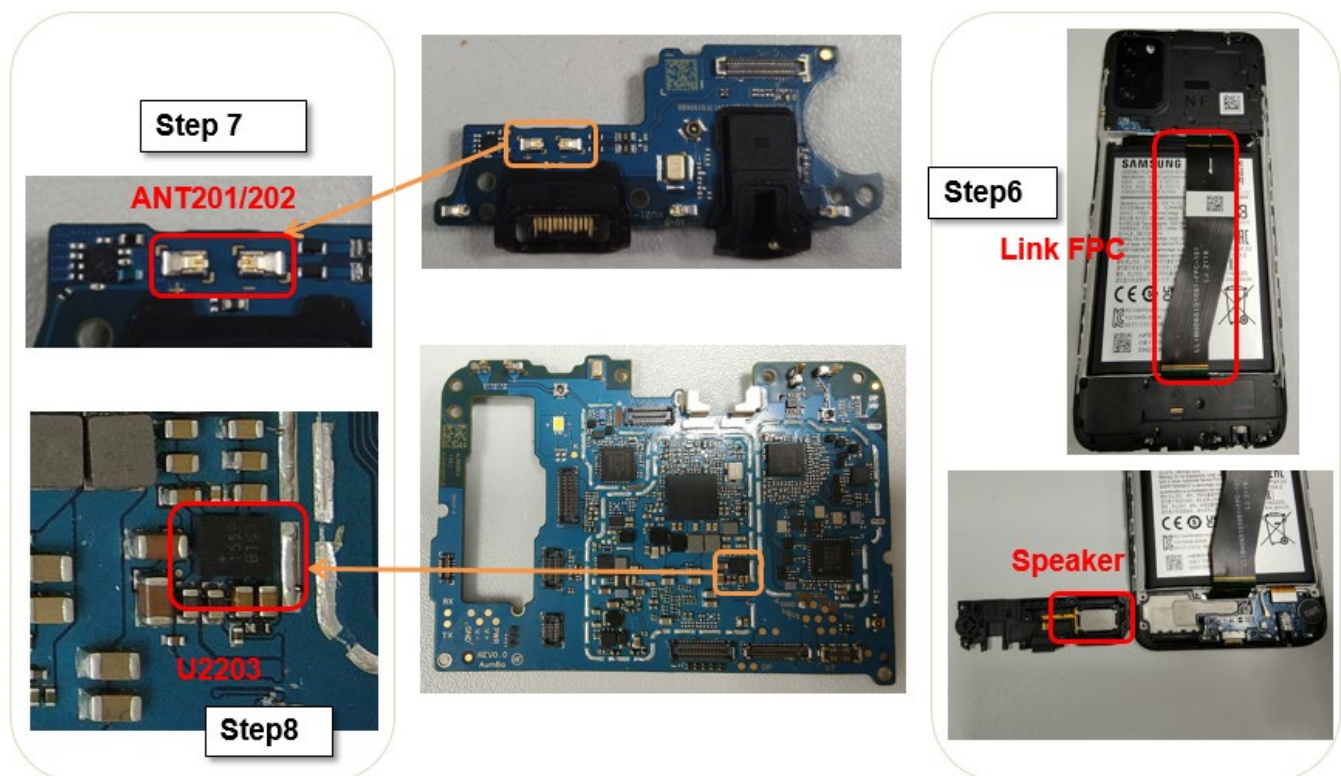
8-3-4. Sound Problem

[illegible]

Step.	Check point	Result value	Defect point
1	Confirm the defect symptom.	-	-
2	*#88*# → speaker	No sound	Go to the next step
		Normal	S/W or Settings
3	Replace the speaker, and also replace the <u>Assy</u> case-rear to use a new speaker tape.	Solved	speaker
		Not solved	Go to the next step
4	Activate the speaker path. (*#0*# → Speaker)	-	-
5	Check C-clips of Sub PBA ANT201,ANT202	normal	Go to step 6
		defect	Replace C-Clip
6	Check the PWM output at ANT201	Yes	Go to step 7
		No	Replace Link FPCB
7	Check the PWM output at ANT201	Yes	Go to step 8
		No	Replace SUB PBA
8	Check the PWM output at U2203	Yes	Replace speaker module
		No	Replace U2203 on MAIN PCB

8. Level 3 Repair

Sound Problem



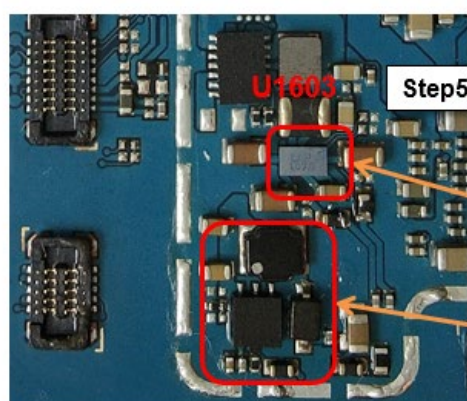
8-3-5. Display Problem

8. Level 3 Repair

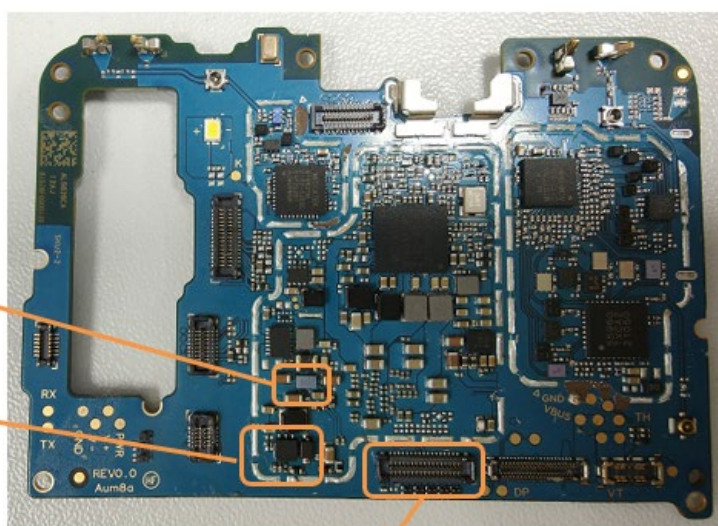
Display Problem

Step	Check point	Result value	Defect point
1	Confirm the defect symptom	-	-
2	Check the LCD connector (J2301)	Broken, dust, corrosion	LCD connector (J2301)
		Loose fitting	Connection
		Normal	Go to the next step
3	Replace the LCD	Solved	LCD
		Not solved	Go to the next step
4	Connect a LCD, and display on with a power supply (power supply voltage : 4.0V)	-	-
5	Check LCM_AVDD = 6.0V (C1616) LCM_AVEE = -6.0V (C1612)	No	Replace the U1603
		Yes	Go to step 6
6	Check LCD_LEDV = 22V (C1617) LCD_LEDK = 0.2(R1608)	No	Replace the U1604, L1603, D1601
		Yes	Replace the Display module

Display Problem



Step6 **U1604**



J2301

Step2

8. Level 3 Repair

8-4. Service Schematics

■ U400_MT6765_BB chip IC , Digital Baseband Processor(Top)

558	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28					
A	NC	WF_IN	WF_IP		EMIO_CA9		NC		EMIO_CS0_N		DVSS		DVSS		DVSS		EMIO_DQ8		EMIO_DQ15		EMIO_DQ31		MSDC0_DAT4		MSDC0_DAT2	MSDC0_DAT7	DVDD1_8_MSD_C0	NC	A				
B	WF_OP	WF_Q	DVSS	EMI_EX_TR	EMIO_CA4		EMIO_CA1	EMIO_CA0	NC		EMIO_DQ21	EMIO_DQ22	EMIO_DQ20		EMIO_DQ4	EMIO_DQ0	EMIO_DQ7		EMIO_DQ13	EMIO_DQ28	DVSS		MSDC0_DAT5	MSDC0_D06_CMD	MSDC0_D01	USB_D	AVDD1_3_USB		B				
C		DVSS	AVDD1_2_VBNC		EMIO_CA6	DVSS	EMIO_CA2		NC		EMIO_DQ18	EMIO_DQ17	DVSS	EMIO_DQ0	EMIO_DQ2	EMIO_DQ1	DVSS	EMIO_DQ11	EMIO_DQ14	DVSS	EMIO_DQ29	EMIO_DQ29	DVSS	MSDC0_DAT3	MSDC0_RSTB	MSDC0_DATA0	USB_D_P		C				
D	BT_IN	BT_IP	DVSS	NC	EMIO_CA5	EMIO_CA3	NC	EMIO_CK0	DVSS	EMIO_DQ16	NC	EMIO_DQ19	DVSS	EMIO_DQ50	EMIO_DQ5	EMIO_DQ51	NC	EMIO_DQ12	EMIO_DQ24	EMIO_DQ27	EMIO_DQ53	DVSS		MSDC0_DAT1	MSDC0_CLK	CHD_D_P	DVSS	AVDD1_8_USB	D				
E			BT_OP	DVSS	EMIO_CA8	EMIO_CA7	EMIO_CK1	EMIO_CS1_N	EMIO_CK_C	EMIO_DQ23		EMIO_DQ52	EMIO_DQ50	EMIO_DQ51	EMIO_DQ6	EMIO_DQ6	EMIO_DQ51	EMIO_DQ9	EMIO_DQ13	EMIO_DQ25	EMIO_DQ29	NC	DVSS	DVSS	CHD_D	CHD_D	CHD_D	AVDD1_2_USB	E				
F	GPS_I	DVSS	BT_QIN	DVSS	DVSS	DVSS	DVSS	DVSS	EMIO_CK_T	DVSS	DVSS	DVSS	DVSS	EMIO_DQ52	DVSS		REF_E_N	DVSS	EMIO_DQ10	DVSS	DVSS	DVSS	DVSS	DVSS	GPIO_E_XT2	GPIO_E_XT1	GPIO_E_XT4	TESTMODE		F			
G	GPS_Q	CONN_WB_PT	HRST_	DVSS			XIN_WBG				DVSS	DVSS	DVSS	DVSS	DVSS	NC				DVSS	DVSS	DVSS	DVSS	GPIO_E_XT0	GPIO_E_XT3	GPIO_E_XT5	DVSS	AUD_C_LK_MIS	DVDD1_8_VBNC	G			
H		AVDD1_8_VBNC	CONN_WF_CT_R12	CONN_WF_CT_R11	CONN_WF_CT_R10					AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	DVSS	DVSS		AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	AVDD2_EMI	GPIO_E_XT6	GPIO_E_XT7	DVSS	RTC32_K_CK	AUD_S_YNC_M		H			
J	DVDD1_8_IORT	CAM_C_LK2	SC16	SDA6	CONN_BT_CLK	CONN_TOP_C_LK																	DVDD1_8_TOP	DVSS	DVSS	DVSS	AUD_C_LK_MO	AUD_S_YNC_M	AUD_D_AT_MIS		J		
K		CAM_P_DN2	CAM_R_ST2	ANT_S_EL2	ANT_S_EL0	ANT_S_EL1	CONN_BT_DA_ATA	DVSS		DVDD1_8_TOP	DVSS	DVSS	DVDD1_8_TOP	DVDD1_8_TOP	DVSS	DVSS	DVDD1_8_TOP	DVSS	DVSS	DVDD1_8_TOP	DVSS	DVSS	DVSS	WATCHDOG	CAM_P_DN3	SRCLK_NA0	DVSS	PWRAP_SPI0_M	PWRAP_SPI0_M		K		
L	CS11A_L1P	CS11A_L1N	CS11A_L0P	CS11A_L0N	CS11B_L0P			DVSS			DVSS		DVDD1_8_TOP	DVSS	DVSS	DVSS	DVDD1_8_TOP	DVSS	DVSS	DVDD1_8_TOP	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	PWRAP_SPI0_M	PWRAP_SPI0_M		L		
M	CS11A_L2P	CS11A_L2N	CS11B_L1P	CS11B_L1N	CS11B_L0N	DVSS	DVSS			DVDD1_8_TOP			DVSS	DVDD1_8_TOP		DVSS	DVSS	DVDD1_8_TOP		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	PWRAP_SPI0_M	PWRAP_SPI0_M		M		
N	CS10A_L1P	CS10A_L1N	CS10A_L0P	CS10A_L0N	CS10B_L2P	DVSS	DVSS			DVSS			DVDD1_8_TOP	DVSS	DVSS	DVSS	DVDD1_8_TOP		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		N	
P	CS10A_L2P	CS10A_L2N	CS10B_L0P	CS10B_L0N	CS10B_L2N	DVSS				DVDD1_8_TOP			DVSS	DVDD1_8_TOP	DVDD1_8_TOP	DVSS	DVSS	DVDD1_8_TOP		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		P	
R	CS12A_L0P	CS12A_L0N	CS10B_L1P	CS10B_L1N	CS10B_L2P					DVDD1_8_VDD_MODE	DVSS		DVSS	DVDD1_8_TOP		DVSS	DVSS	DVDD1_8_TOP		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		R	
T	CS12A_L1P	CS12A_L1N	CS12A_L2P	CS12A_L2N	CS12B_L1N			DVSS		DVDD1_8_VDD_MODE	DVSS		DVSS	DVDD1_8_VDD_MODE	AVDD1_8_PLL2_P	AVDD1_8_PLL2_P	DVSS	DVDD1_8_TOP		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		T	
U	DVDD1_8_IORB	AVDD1_2_CSI	CS12B_L0P	CS12B_L0N		SC14				DVDD1_8_VDD_MODE	DVSS		DVSS	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVSS	DVSS	DVDD1_8_VDD_MODE		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		U	
V	CAM_R_ST1	CAM_R_ST0	SDA2	SCL2		SDA4		DVSS		DVDD1_8_VDD_MODE	DVSS		DVSS	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVSS	DVSS	DVDD1_8_VDD_MODE		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		V	
W		CAM_C_LK1	EINT10	EINT9	CAM_C_LK0	CAM_P_DN0	EINT11	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		W	
Y	SRCLK_NA1	CAM_P_DN1	EINT7	EINT6		EINT4	EINT5	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVDD1_8_VDD_MODE	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		Y	
AA	KPRO_W0	UTXD0	URXD0	EINT3	EINT2	EINT8	EINT1			DVSS			DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		AA	
AB	KPRO_W1	I_0	KPCOL_0	EINT0	SCL0	CDM5P_SA				DVSS			DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		AB	
AC	SPI0_C_SB	SPI0_M_O	PWM0	SDA0	CDM3P_SA	BPI_BU_S7			REF_E_T0_B	APC							AVDD1_8_CPU	GPIO_E_XT15	GPIO_E_XT14	GPIO_E_XT13	GPIO_E_XT12	GPIO_E_XT11	GPIO_E_XT10	GPIO_E_XT9	GPIO_E_XT8	GPIO_E_XT7	GPIO_E_XT6	GPIO_E_XT5	GPIO_E_XT4	GPIO_E_XT3		AC	
AD	SPI0_M_I	SPI0_C_LK	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO		AD
AE	SCL1	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO		AE
AF	SDA1	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO		AF
AG	NC	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO	BPI_PA_VMO		AG
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28					